My teaching and mentoring experiences have shown me that students learn most effectively when they actively engage with course material, work through problems with others, and connect concepts to real questions. I see my role as an instructor as creating a clear and supportive structure for that engagement by encouraging students to ask questions, explain their reasoning, and stay curious about how chemistry connects to the world around them. My goal is to help students build a strong foundation in chemistry while developing problem-solving and communication skills they can carry into future courses, research experiences, and careers.

I put this philosophy into practice by creating opportunities for students to work through problems rather than simply receiving answers. As a teaching assistant for Organic Chemistry and Principles of Chemistry laboratories, I used activities such as safety scenarios and pre-lab concept discussions to help students prepare for experiments, identify important safety considerations, and connect laboratory procedures to the underlying chemistry. During lab, I asked students to work in small groups and moved from group to group to check their progress, ask questions, and address misconceptions. Rather than immediately giving an answer when students encountered a problem, I encouraged them to explain their reasoning and consider possible next steps. This approach allowed me to identify where students were struggling while helping them become more confident in solving problems independently.

"Sharear made frequent rounds to each and every student during lab, checking on their progress and ensuring they understood the process." - Student feedback

My teaching and mentoring experiences have also shown me that students enter a course with different levels of preparation, prior knowledge, confidence, academic goals, communication styles, and familiarity with college-level science. For me, supporting students from diverse backgrounds and perspectives begins with recognizing these differences without lowering expectations. I therefore try to make expectations clear and provide multiple ways for students to engage with difficult concepts, including diagrams, real-world examples, guided questions, and discussion with peers. I also value frequent, low-stakes opportunities for students to check their understanding before major assessments. This commitment to accessible learning led me to develop RNA Explorer, a browser-based tool that allows students to examine RNA structures without installing specialized software or relying on high-end computing resources. For me, accessibility in teaching means reducing unnecessary barriers so that more students can engage directly with scientific ideas while still being challenged to explain and apply what they learn.

My research mentoring has reinforced this approach to teaching. I have worked with students from high school through graduate levels, including undergraduates who began with limited experience in computational chemistry and programming. By breaking projects into manageable steps, working through problems together, and gradually giving students greater independence, I helped an undergraduate researcher complete a thesis and become a co-author on a peer-reviewed publication. My commitment to undergraduate mentoring was also recognized with an Exceptional Mentorship Award at the Fifth Annual Research Symposium at Fayetteville State University. These experiences have strengthened my belief that students do not need to arrive with the same preparation to contribute meaningfully; with clear guidance, opportunities to practice, and increasing responsibility, students can develop both competence and ownership of their learning.

I continue to develop my teaching through formal pedagogical training and classroom experience. During graduate school, I completed the Certificate in University Teaching Skills program, where I received training in course design, student engagement, assessment, and instructional technology. In Fall 2025, I also served as a Learning Assistant for a general education course at Penn State,

where I helped organize course materials, facilitated in-class and online group discussions, and occasionally served as a substitute lecturer. This experience expanded my preparation beyond laboratory instruction and gave me additional experience facilitating both in-person and online learning activities for students with different academic backgrounds. In October 2026, I completed the Course in College Teaching (CCT) at the Schreyer Institute for Teaching Excellence at Pennsylvania State University. This professional development strengthened my approaches to course design, assessment, active learning, and reflective teaching practice.

In my future teaching, I will build on my emphasis on active engagement, clear expectations, accessible participation, and increasing student independence by using course structures that give students frequent opportunities to practice and allow me to respond when concepts need to be revisited. I plan to combine higher-stakes assessments with short quizzes, in-class polling, and group problem solving so that students can check their understanding before major evaluations. I will also use structured peer-learning activities, including approaches inspired by Peer-Led Team Learning (PLTL), in which students explain their reasoning to one another and work toward solutions collaboratively. To make participation more accessible, I will provide individual thinking time before small-group and whole-class discussion so that students have multiple ways to contribute. In laboratory and research settings, I will extend the same approach by asking students to connect chemical principles to observations, data, and decisions they make while investigating a problem. Whether teaching face-to-face or online, my goal will be to maintain a clear course structure, active engagement, and regular feedback while adapting activities to the format of the course. I also want students to develop responsible habits when using emerging tools such as generative AI by evaluating the accuracy of generated material and being able to explain the underlying scientific reasoning themselves.

My teaching, mentoring, and research experiences prepare me to contribute across the chemistry curriculum, including general chemistry, biochemistry, and related laboratory courses. In biochemistry and related courses, I would connect molecular-level concepts such as intermolecular interactions, thermodynamics, structure, and molecular recognition to biological function. I would also incorporate authentic biochemical and structural datasets through selected assignments, case studies, and short projects. Students could examine biomolecular structures, interpret experimental measurements, and use accessible computational tools to analyze and visualize data. As opportunities arise, I would be interested in developing an upper-level or special-topics course in biophysical and computational approaches to biomolecular structure and function.

Across my courses, my goal is to help students build a strong foundation in chemistry and biochemistry while developing the confidence to work through unfamiliar problems, communicate their reasoning, and collaborate with others. I hope to create a learning environment in which classroom instruction, laboratory and computational experiences, and undergraduate research reinforce one another. I want students with different levels of prior preparation and varied professional interests to see a meaningful place for themselves in chemistry and biochemistry and to leave my courses better prepared for advanced study, research, professional education, or careers in science and related fields.